Exercise 3: SQL Fundamentals

1. SQL Joins

```
SELECT EmployeeID,
       firstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	firstName	LastName	Department	Salary	Pro-ID	Pro Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100000	Completed
1	John	Doe	IT	70 000	103	cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	ongoing
3	Bob	Johnson	Finance	75 000	104	financial Analysis	90 000	ongoing
5	Emma	Wilson	Sales	65000	105	market Expansion	65000	Completed
6	Michael	Clerk	finance	80 000	206	Risk Management	80 000	Pending

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2. SELECT EmployeeID,
first_name,
Last_name,
Department,
Salary,
ProjectID,
ProjectName,
Budget,
Status

FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;

EmployeeID	firstName	lastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100 000	Completed
1	John	Doe	IT	70000	103	Cybersecurity audit	75 000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial analysis	90 000	Ongoing
4	David	Brown	IT	72000	null	null	null	null
5	Emma	Wilson	Sales	65000	105	market Expansion	65 000	Completed
6	Michael	Clark	Finance	80000	106	risk management	80 000	Pending

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(3)- SELECT Project ID,
Project Name,
Budget,
Status,
EmployeeID,
FirstName,
LastName,
Department,
Salary
FROM Employees AS A
RIGHT JOIN Projects AS B
~~ON A.Employees = B.Pro~~
ON A.EmployeeID = B.EmployeeID;

Project ID	Project Name	Budget	Status	EmployeeID	FirstName	LastNam	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70 000
102	training Employee	50000	Ongoing	2	Alice	Smith	HR	60 000
103	cybersecurity Audit	75 000	Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75 000
105	market Expansion	65000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk management	80000	Pending	6	Michael	Clark	Finance	80 000

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⑥ SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary,
ProjectID,
ProjectName,
Budget,
Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID

EmployeeID	First Name	Last Name	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	Null	null	null	null
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

⑦ ~~Employees~~
SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary
FROM Employees
WHERE Salary > 70 000;

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EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

⑧
```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary.
FROM Employees
WHERE Department = 'IT' OR Department = 'finance';
```

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

⑨
```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM Projects
WHERE Status != 'completed';
```

ProjectID	ProjectName	Budget	Status
102	Employee Training	50 000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

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(10) SELECT ProjectID,
Project Name,
Budget,
Status
FROM Projects
WHERE Budget > 70000 AND Status != 'completed';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

(11) SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary,
City
FROM Employees
WHERE City = 'New York' OR City = 'Toronto'
ORDER By Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Mianaer	Clark	Finance	80 000	New York
1	John	Doe	IT	70 000	New York
3	Bob	Johnson	Finance	75000	Toronto

(12) SELECT EmployeeID,
First Name,
LastName,
Department,
Salary
FROM Employees
ORDER BY Salary DESC LIMIT 3;

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EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	finance	80 000
3	Bob	Johnson	finance	75000
4	David	Brown	IT	72000

(13) SELECT Department,
SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;

Department	TotalSalary
finance	155 000
IT	142 000
HR	60 000
Sales	65000

(14) SELECT City,
AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
~~HAVING AVG(Salary) > 65000;~~
HAVING Average(Salary) > 65000;

City	AverageSalary
New York	75 000
Toronto	75000
London	72 000

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(15) SELECT Department,
COUNT (EmployeeID) AS EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT (EmployeeID) > 1;

Department	EmployeeCount
IT	2
Finance	2

(16) SELECT status,
COUNT (ProjectID) AS ProjectCount
FROM Projects
GROUP BY status
HAVING COUNT (ProjectID) >= 2;

Status	ProjectCount
Ongoing	2
Pending	2

(17) SELECT EmployeeID,
FirstName,
LastName,
SUM (Budget) AS TotalProjectBudget
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
GROUP BY EmployeeID,
FirstName,
LastName
HAVING SUM (Budget) > 150 000;

EmployeeID	FirstName	LastName	TotalProjectBudget ~~total budget for project~~
1	John	Doe	175000

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